

SMARTER AMI-SP

-AI/ML Modelling
-Data Analytics
-NILM

SWOT – AMI(SP) - India

1. **Consumer acceptance and Mass Roll** outs @250 Million smart meters ,MoP GoI.
2. Credibility ,Transparency ,Dispute-less - **costs/billing services** for Utility
3. **True consumption , credit and recharge values ,** easy to monitor and understand by the consumer
4. **Real Time values** of consumption , credit and balance , which are **free of ambiguity and dispute**
5. **Licensee's to Follow the codes** laid down by the **regulator**
6. Introduction of a **universal HES/MDM** solution and **Interoperable Systems/ Technology**
7. **Integration** of MDM services with **legacy billing / ERP software** of the Discom's
8. Reduction in **AT&C losses** and True Audits
9. **Load Forecasting T++**

Challenges

DISCOMs and AMI-SP's Vs the CONSUMER sentiment/thought/benefit

- **Ramping AMI-SP Pre Paid ROLL OUT ?**
- **Improve consumer ENGAGEMENT ?**
- **Enhance consumer CONFIDENCE ?**
- **Elevate consumer EXPERIENCE ?**
- **Invoke consumer PARTICIPATION ?**

SaaS Platforms - CONSUMER centric and friendly

Opportunity

Big Data Capture and Ingestion @15-30 Min Int – Granular 90 Sec Intervals

- **350-500 KB/Meter/ Month**
- **5 MB/Meter/ Year**
- **1250 TB to approx. 6500 TB /Year - 250 Mil Nodes**

AI @ Edge – Data Analytics

1. Identify **power hungry devices** , their inefficient usage and assist in their regulated operation
2. **Predict consumption/demand** individual and aggregated **based on historical data** and **weather patterns** etc for more efficient use of the energy resource
3. Future proof the system **adoption for ToU /ToD** tarrif's **Vs Cost Peak Power - energy exchanges**
4. **Switch between multiple sources** of energy -renewables and others like **V2G** vice versa etc.
5. **Voluntary Participation** by consumers in **Demand Response** via load adjustments

- ✓ **Build Edge Intelligence @ the Meter NIC level**
- ✓ **Optimize the usage of the cloud**

Intrusive Device -Approach

Demand Side Management (DSM) measures in the Energy Sector is a cost-effective tool. As a customer strategy, DSM programs encourage the installation of end-use technologies that consume less energy, thereby reducing and/or shifting the customers' overall electric bill. DSM programs can help utilities to reduce their peak power purchases on the wholesale market, thereby lowering their overall cost of operations. The capacity building and other support is essential for the DISCOMs to implement DSM in their respective areas.

Device Approach – Need for physical sensors on the appliances

NILM Approach

- Data-driven approach that relies on near real-time parameters captured from smart meters

- Enables detailed energy usage analysis without the need for intrusive sensors

NILM- Raw Data Acquisition

1. Voltage
2. Current
3. Power Factor
4. Power Off / On Cycle
5. Energy Cycle

Data is securely ingested into platform via MQTT protocols and stored in both time-series NoSQL (MongoDB) and relational (MySQL) database structures.

NILM- Data Cleansing

1. Singal smoothening
2. Time Stamp Alignment
3. Phase wise energy (if 1ph/3ph loads)
4. Power Consumption – Granularity & higher Sampling Frequency

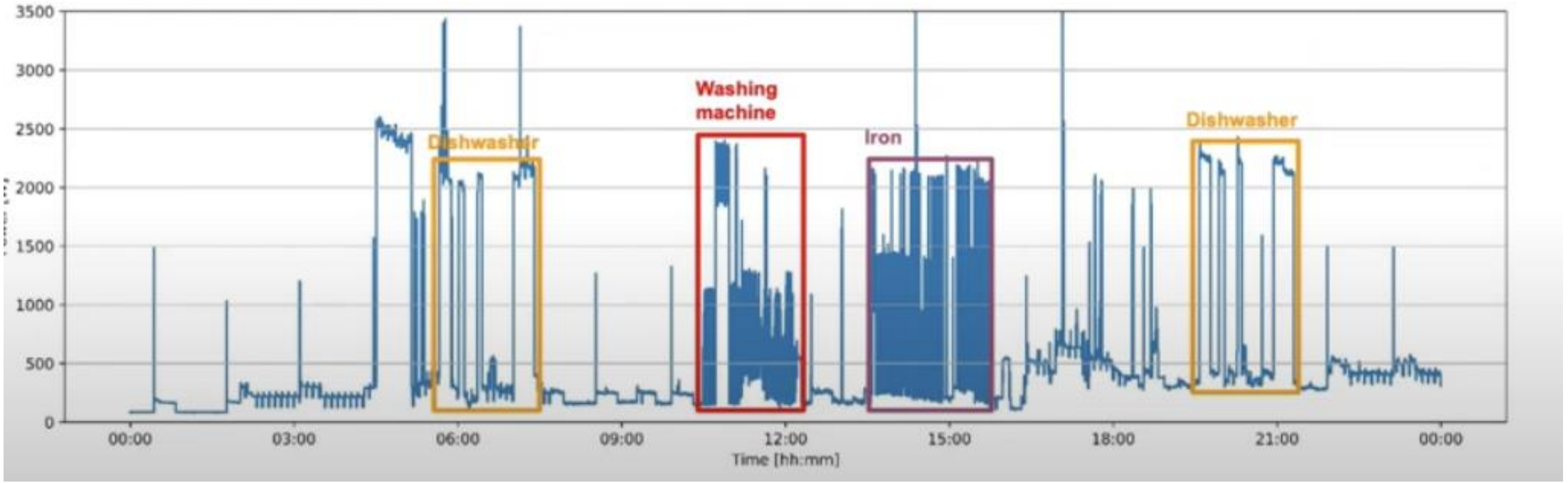
Key parameters are extracted - load transients, step changes, and load signatures (determine set patterns) from the power consumption waveform using statistical inference.

NILM- AI/ML engine

1. Supervised Pattern Recognitions – Device Signature Matching
2. Un-Supervised Clustering – Use K-Mean & DBSCAN Algorithms
3. Edge Detection – Device switching on active/reactive power changes
4. **Real Time comparison with data sets and continuous self learning**

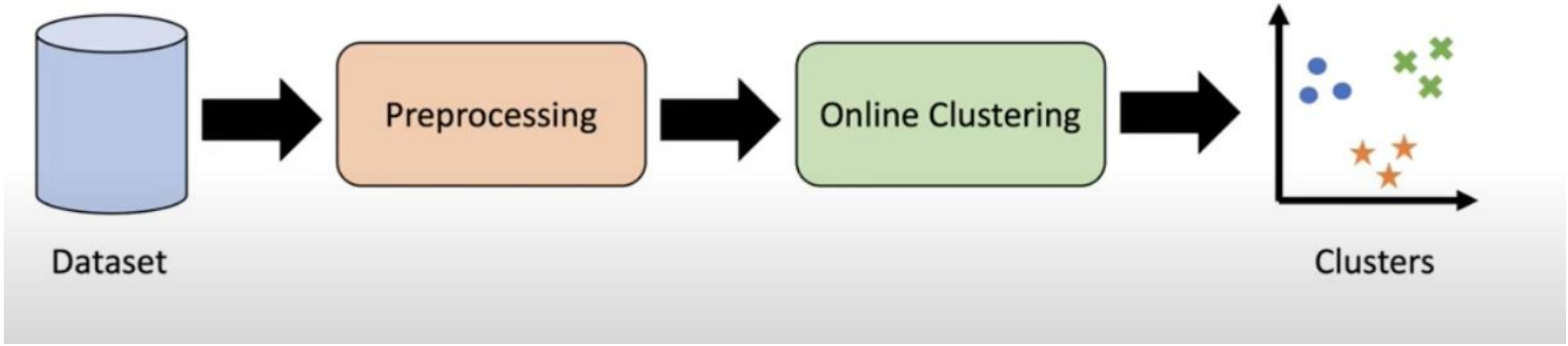
The models are trained using historical datasets and continuously validated with labelled device datasets during onboarding to ensure accuracy.

NILM – Pattern Recognition & Learning



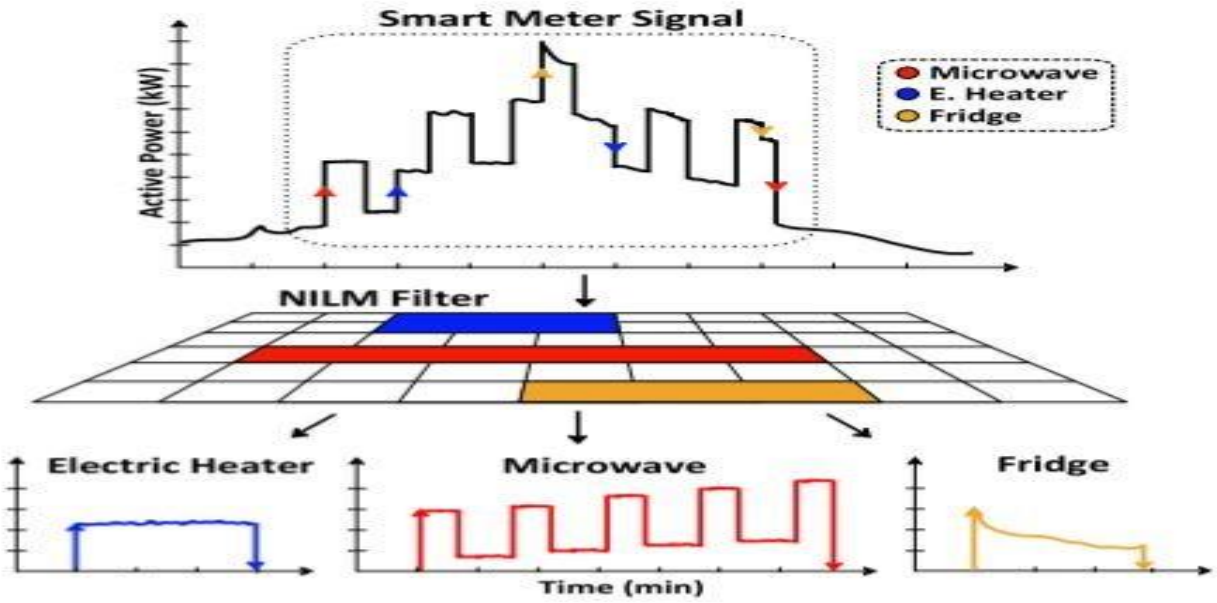
Based on the **DIGITAL SIGNATURE** of the devices Vs data set

NILM – Clustering and Probability



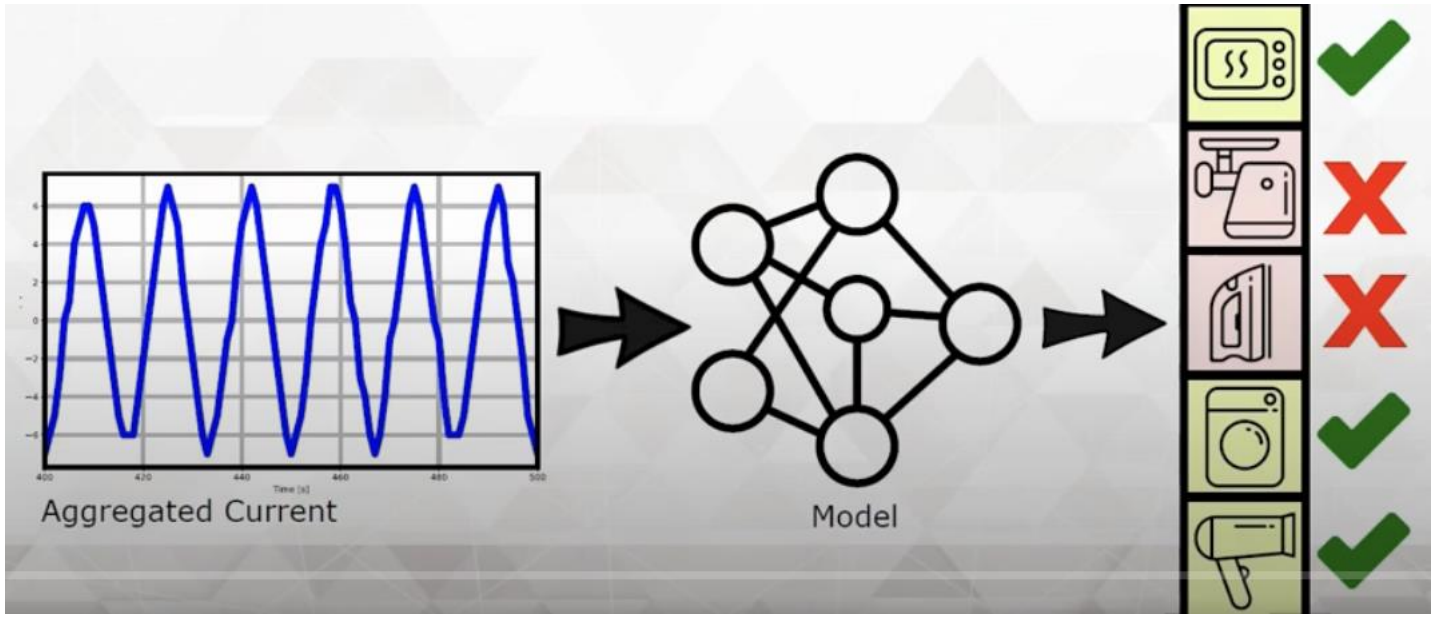
To over come variation /anomalies of data sets(Library) and to bring in more rationalised data the AI/ML model uses the probability & clustering model for better analytics

NILM – Modelling and Filtering



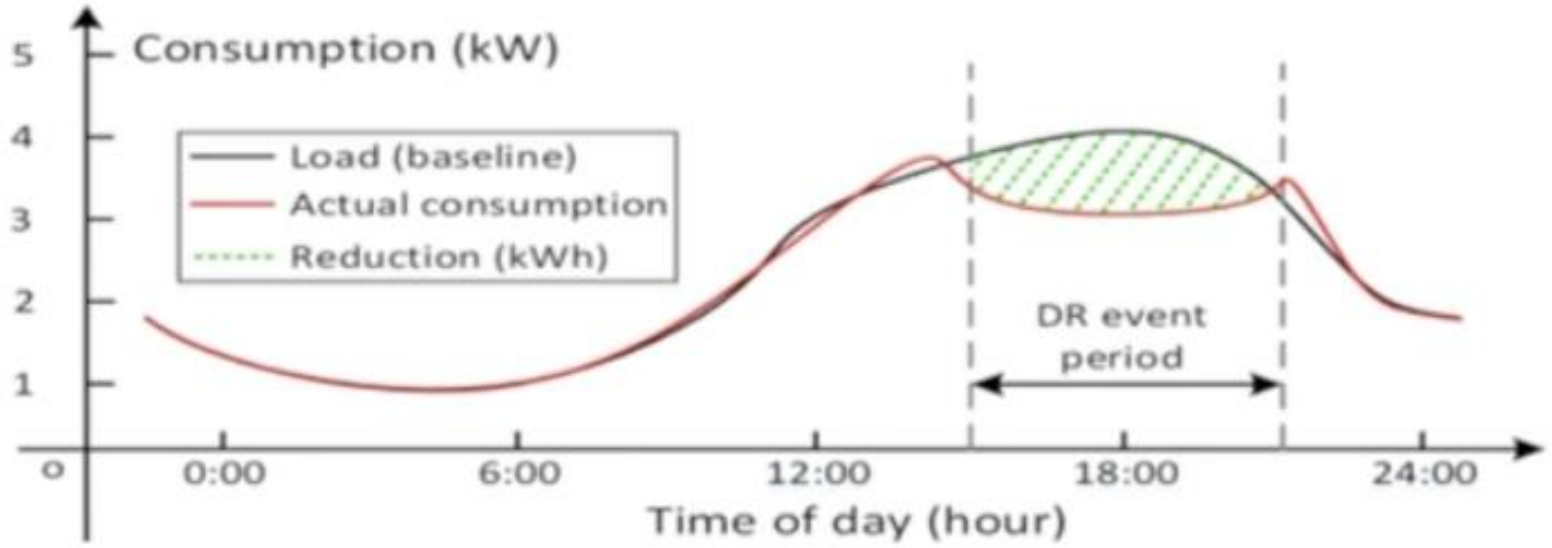
This stage of NILM filtering technique through the AI/ML engines for the data being provided in real time from the energy meter helps identifying the devices connected and operational in the circuit

NILM – Final Output Consumer Dashboard



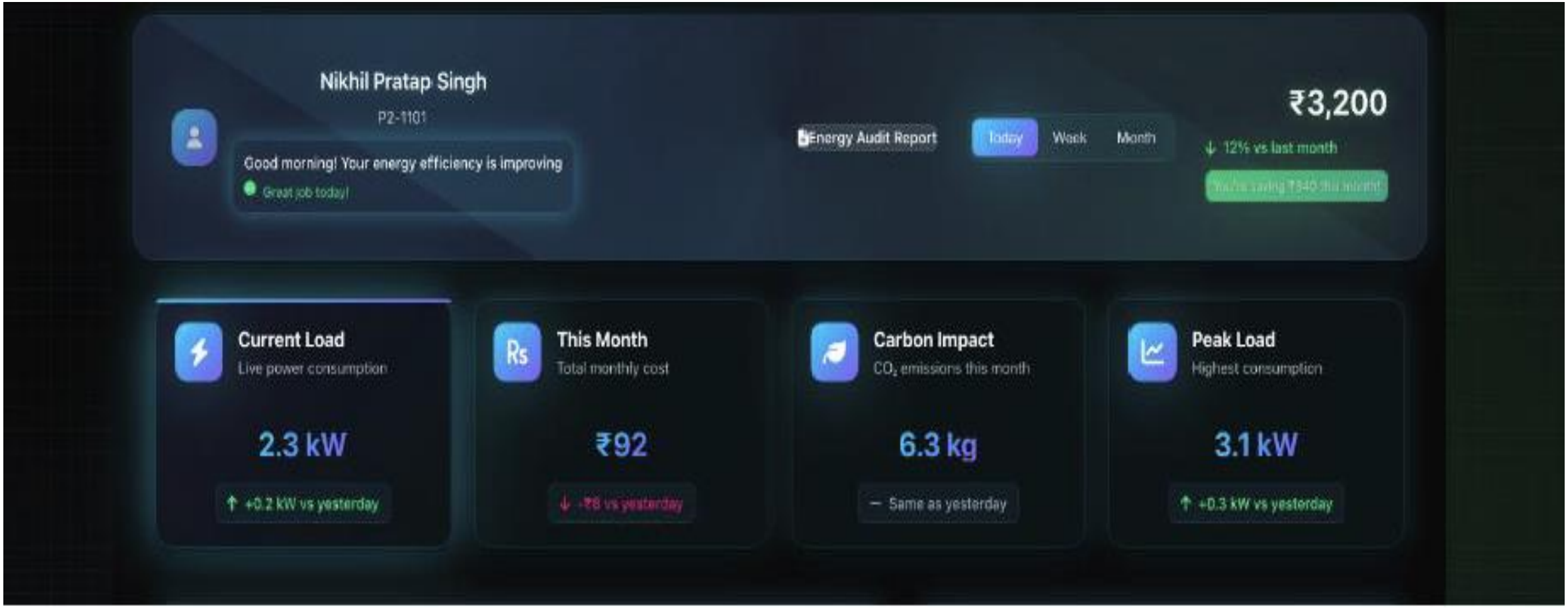
NILM model is able to correctly predict the device in use in real time and it’s contribution to the total energy consumption

NILM – Use case – Demand-side Mngmt



The **Utility and the consumer** on a real time basis , **can play collaborative and participative roles** where in ToU & ToD pricing mechanism can be offered to incentivise the consumers for energy spend and also help Utilities to regulate and **create predictability** in their **demand/supply** bringing **fiscal prudence** and efficiently **balancing between** the use of **renewable and non-renewable sources** of energy as alternate sources of supply in the coming time

Live PoC in PvvNL _NPCL Consumers - Mobile dashboard snap shot



Insights - Actionable +ve Outcomes

- ✓ Patterns - Previous Vs Present Vs Predicted consumption
- ✓ Carbon impact (block chain token / credit generation as tangible tradeable commodities)
- ✓ Savings

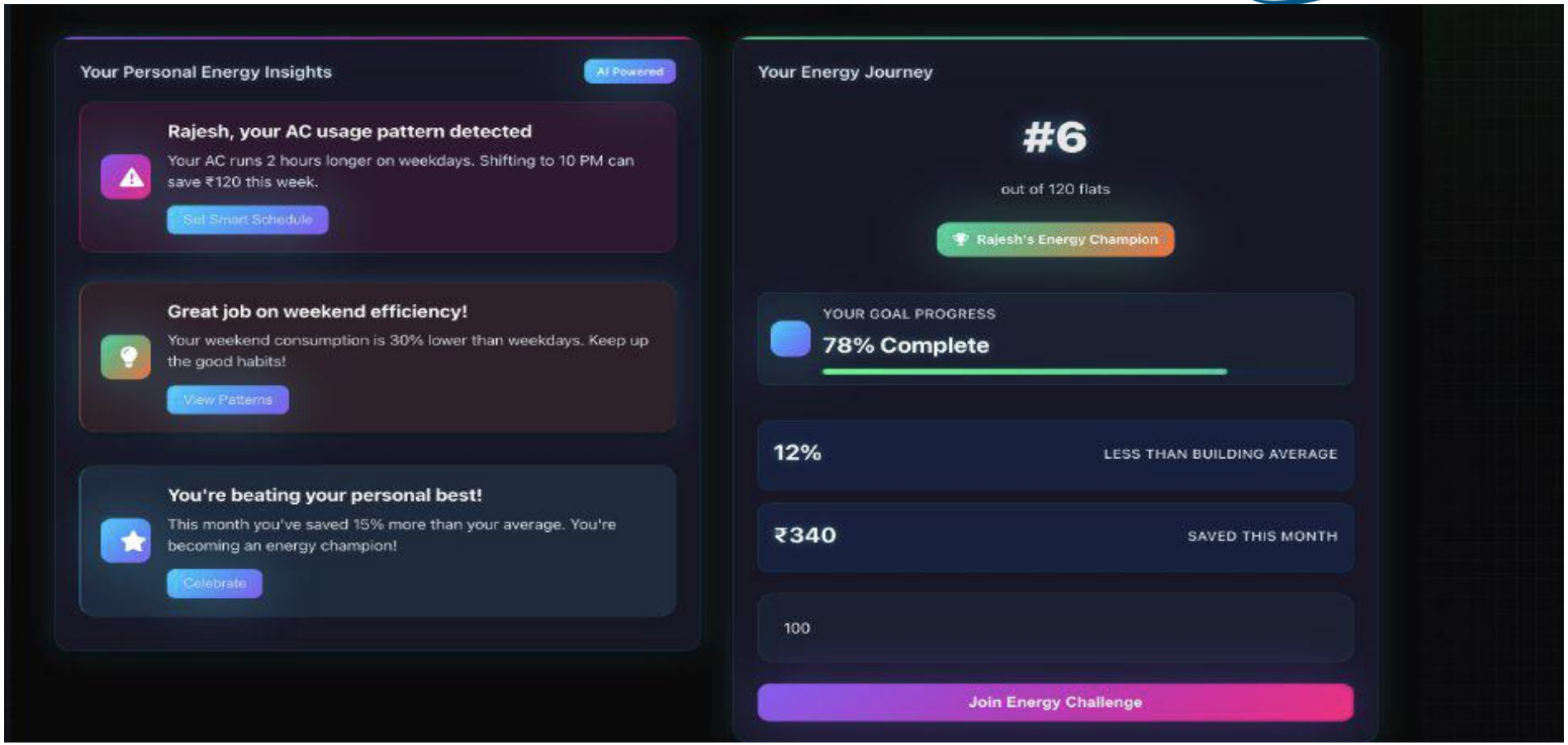
AI/ML Platform generated Analytics- Mobile App snap shot



Insights – Actionable for consumer participation

- ✓ Household device consumption and contribution breakdown –value(Kw/Rs) & %
- ✓ Regulate and optimize device usage / timing etc

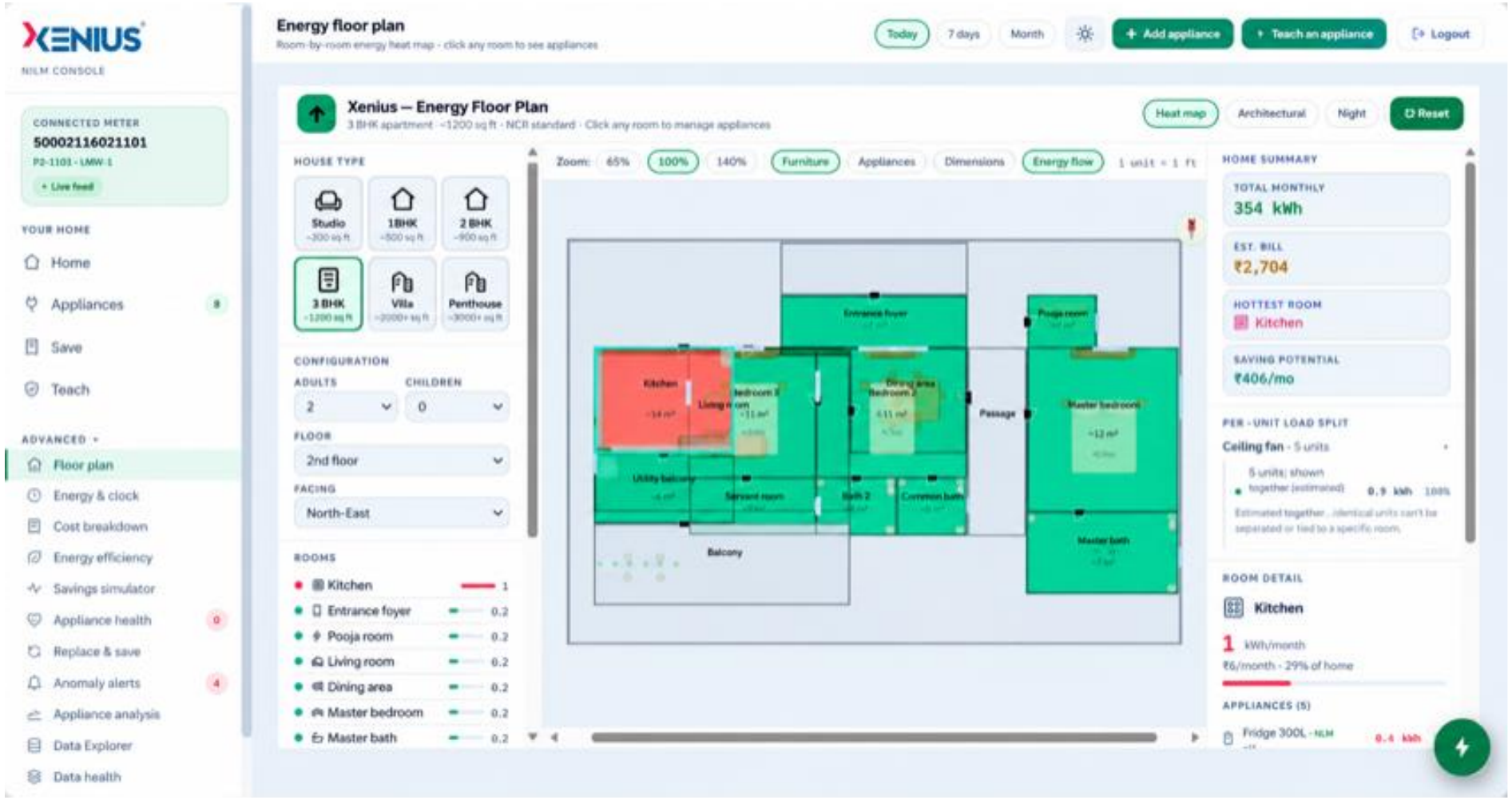
AI/ML Platform generated Analytics- Mobile App snap shot



Insights – Applauding and Appreciation the consumer

- ✓ Posts savings in Rs INR
- ✓ Prompts Target Vs Goals accomplishment with peers too
- ✓ Suggestive advise on device usage optimization

NILM – Digital Twin Heat Map
Home Foot Print



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Humanizing Machines

Jaideep SINGH
Advisor : Strategy and Stake Holde Engagement